



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ D ₃ – ⁵ F ₄ ^o	248.7920	248.7169	1204.9924	–	–	–	–	first-step line of the in-gas-cell laser resonance ioniza...
a ¹ D ₂ – ³ D ₃ ^o	270.3200	270.2399	1109.0280	2.030	6.040e+07	–	–	saturation spectroscopy in a hollow-cathode discharge: is...
³ F ₄ – ³ D ₃ ^o	270.6696	270.5894	1107.5954	1.692	4.180e+07	–	–	saturation spectroscopy in a hollow-cathode discharge: is...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁹⁵Pt hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
³ D ₃	5702.64(23)	–	F=5/2: -11405.28 F=7/2: +8553.96
³ D ₃ ^o	1917.2(9)	–	F=5/2: -3834.40 F=7/2: +2875.80
⁵ F ₄ ^o	1150(110)	–	F=7/2: -2875.00 F=9/2: +2300.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
³ D ₃ – ⁵ F ₄ ^o	248.792	198-194	2430(230)
³ D ₃ – ⁵ F ₄ ^o	248.792	196-194	1390(270)
³ D ₃ – ⁵ F ₄ ^o	248.792	192-194	-1430(270)

Shift = ν(first isotope) – ν(second isotope).